

The Department of Chemical Engineering is a part of the Faculty of Engineering and Architecture and is fully accredited by the Pakistan Engineering Council. Its core engineering focus is on the transformation of raw feedstock into valuable commercial products, such as transport fuels, agro-industry fertilizers, processed foods, clean water, and construction cement. The curriculum encompasses chemical process and equipment designs, cost estimation, process control, optimization, and the safe scaling of pilot projects to industrial operations. The department was founded in September 2005 with only 4 faculty members and 14 students at the Takatu Campus, and today it has grown to 21 faculty members, most of whom are foreign-qualified. It is equipped with state-of-the-art research laboratories that are actively engaged in cutting-edge research covering nanotechnology, algal cultivation, biomass and bioenergy, waste management for product recovery, wastewater treatment, and polymer composites. The vision of the department is to be among the leading Chemical Engineering Departments of the country, with programs providing quality graduates for the chemical and allied industry, focusing on research and development. Its mission is committed to establishing a conducive environment for professional education and research in the field of Chemical Engineering to address the needs of the country and region. The Program Educational Objectives target producing chemical engineers who can display specific traits within a few years of graduation, which include professional competency in chemical engineering to address the needs of the chemical and allied industries as well as the research and development sector, contributing to the solution of industrial problems and technological and societal issues, and pursuing professional development and technical achievement through continuing education.

For the Bachelor of Science in Chemical Engineering program, the admission requirements dictate an academic background of F.Sc Pre-Engineering from a recognized board or an equivalent certification with at least 60% marks, or a Diploma of Associate Engineer in a relevant field with a minimum of 60% marks. The degree requirements involve completing 136 total credit hours across 45 to 49 total courses, while maintaining a minimum academic standing of a CGPA of 2.0 or higher. The curriculum plans and program schema by semester outline two distinct course pathways based on enrollment intake years, where credit hour structures denote theory plus lab hours. Track A is for student intake terms from Fall 2019 to Fall 2021. The first semester involves Islamic Studies or Ethics for 2 plus 0, Pakistan Studies for 2 plus 0, Introduction to Computers for 1 plus 1, Engineering Drawing for 0 plus 1, Calculus and Analytical Geometry for 3 plus 0, Introduction to Chemical Engineering for 2 plus 0, and Organic and Inorganic Chemistry for 2 plus 1. The second semester requires Applied Physics for 3 plus 1, Computer Aided Engineering Drawing for 0 plus 1, Chemical Process Calculations-I for 3 plus 0, Linear Algebra and Differential Equations for 3 plus 0, Reading and Writing Skills for 3 plus 0, Physical and Analytical Chemistry for 2 plus 1, and Workshop Practice for 0 plus 1. The third semester includes Programming Fundamentals for 2 plus 1, Electrical Technology and Electronics for 2 plus 0, Chemical Engineering Thermodynamics-I for 3 plus 1, Particle Technology for 3 plus 1, and Fluid Mechanics-I for 3 plus 1. The fourth semester consists of Chemical Process Calculations-II for 3 plus 0, Applied Mathematics in Chemical Engineering for 3 plus 0, Communication and Presentation Skills for 3 plus 0, Chemical Process Technology for 2 plus 1, Chemical Engineering Thermodynamics-II for 3 plus 0, and Mass Transfer for 3 plus 0. The fifth semester covers Separation Processes for 2 plus 1, Fluid Mechanics-II for 3 plus 1, Numerical Analysis and Software Applications for 2 plus 1, Probability and Statistics for 2 plus 0, Heat Transfer for 3 plus 1, and Engineering Mechanics for 2 plus 0. The sixth semester includes Fuels and Energy for 3 plus 1, Chemical Reaction Engineering for 3 plus 1, Simultaneous Heat and Mass Transfer for 2 plus 1, Engineering Materials for 3 plus 0, Technical Report Writing for

2 plus 0, and Engineering Economics for 2 plus 0. The seventh semester involves Logical and Critical Thinking for 2 plus 0, Chemical Plant Design for 3 plus 0, Chemical Process Design and Simulation for 2 plus 1, Project-I for 0 plus 2, Transport Phenomena for 3 plus 0, and Instrumentation and Process Control for 3 plus 1. The eighth semester completes Track A with Industrial Management for 3 plus 0, Elective-I for 2 plus 0, Elective-II for 2 plus 0, Entrepreneurship for 2 plus 0, Maintenance Engineering and Risk Management for 2 plus 0, and Chemical Engineering Plant Design Project for 0 plus 4.

Track B is designed for student intake terms from Fall 2022 onwards. The first semester covers Islamic Studies or Ethics for 2 plus 0, Reading and Writing Skills for 3 plus 0, Calculus and Analytical Geometry for 3 plus 0, Physical and Analytical Chemistry for 2 plus 1, Introduction to Computer and Programming Concepts for 2 plus 1, Introduction to Chemical Engineering for 2 plus 0, and Engineering Drawing and Graphics for 1 plus 1. The second semester includes Pakistan Studies for 2 plus 0, Applied Physics for 3 plus 1, Linear Algebra and Differential Equation for 3 plus 0, Chemical Process Calculations-I for 2 plus 0, Chemical Process Industries for 2 plus 0, Organic and Inorganic Chemistry for 2 plus 1, and Workshop Practice for 0 plus 1. The third semester entails Particle Technology for 3 plus 1, Chemical Engineering Thermodynamics for 3 plus 1, Communication and Presentation Skills for 3 plus 0, Applied Mathematics in Chemical Engineering for 3 plus 0, and Fluid Mechanics-I for 3 plus 1. The fourth semester consists of Numerical Methods for Chemical Engineers for 2 plus 1, Chemical Process Calculations-II for 3 plus 0, Logical and Critical Thinking for 2 plus 0, Electrical Technology and Electronics for 2 plus 0, Engineering Mechanics for 2 plus 0, and Heat Transfer-I for 3 plus 1. The fifth semester encompasses Engineering Material for 3 plus 0, Fluid Mechanics-II for 3 plus 1, Heat Transfer-II for 3 plus 1, Mass Transfer for 3 plus 1, and Probability and Statistics for 2 plus 0. The sixth semester features Technical Report Writing for 2 plus 0, Engineering Economics for 2 plus 0, Simultaneous Heat and Mass Transfer for 3 plus 1, Chemical Reaction Engineering for 3 plus 1, Fuels and Energy for 3 plus 1, and Maintenance Engineering and Risk Management for 2 plus 0. The seventh semester requires Instrumentation and Process Control for 3 plus 1, Chemical Plant Design for 3 plus 0, Elective-I for 3 plus 0, Process Modeling and Simulation for 2 plus 1, and Project-I for 0 plus 2. The eighth semester finalizes the track with Transport Phenomena for 3 plus 0, Elective-II for 3 plus 0, Biochemical Engineering for 2 plus 0, Industrial Management for 3 plus 0, Entrepreneurship for 2 plus 0, and Project-II for 0 plus 4.

For the Master of Science in Chemical Engineering program, the admission requirements mandate an academic background of 16 years of education or equivalent, such as a 4-year bachelor's degree in Chemical Engineering or Petroleum and Gas Engineering from a recognized University. The academic minimums require at least 60% marks under the annual grading setup, or a program CGPA of 2.5 out of 4.0. Additionally, standardized testing is required, necessitating a minimum score of 50% marks on the GAT General assessment test. The degree requirements for the MS program consist of a total of 30 credit hours and 42 to 44 total listed courses, although this standard catalog total traditionally refers to baseline credit options, as typical graduate tracks complete 8 to 10 actual semester courses. Finally, students must maintain a minimum academic standing with a CGPA of 2.5 or higher.